

ADVANCED WATER SAFETY

Health & Physical Education | Year 9-10 | Scaffolded Edition

Word banks, sentence starters, argument frameworks,
and calculation guides throughout.
Same content as the standard workbook.
All 7 sections covered.

STUDENT NAME

CLASS

TEACHER

How the scaffolds work

- Word banks: key terms to choose from — you do not have to think of them yourself
- Sentence starters: begin the sentence for you — just finish it in your own words
- Argument frameworks: break an argument into parts (claim, evidence, explanation)
- Calculation guides: show you the formula and what numbers to substitute in

1 Advanced Data Analysis — Victorian & Global Drowning Trends**Key Data — Keep This Handy**

- 54 fatal drownings Victoria 2023-24. Population approx. 6.9 million.
- Rate formula: deaths / (population / 100,000). So: $54 / 69 = 0.78$ per 100,000.
- Australian national rate = 1.2 per 100,000 (RLSSA). Victoria is BELOW average.
- CALD communities = ~40% of Victorian drowning deaths, ~28% of population.
- 85% of Victorian drowning deaths (46 of 54) were male.
- All 54 deaths at unpatrolled locations. Zero at patrolled beaches or pools.

Activity 1.1 — Calculations & Interpretation

~15 min

Q1.1a — Calculation [2 marks]

Calculate the percentage of Victorian drowning deaths that were male.

*Working: $46 / 54 \times 100 = \underline{\hspace{2cm}}\%$ (vs global ~80%)***Word Bank:** over-represented · structural · swim lessons · language barriers · fishing risk · 40% > 28%**Q1.1c — CALD Analysis** [3 marks]

Are CALD Victorians over-represented or under-represented? Explain TWO structural reasons.

*CALD Victorians are because 40% of deaths > 28% population.***Reason 1 — Argument Framework:****Reason:****Because:****This means:****Reason 2 — Argument Framework:****Reason:****Because:****This means:**

2 Physiology of Drowning — What Happens in the Body

Key Facts

- 4-6 minutes without oxygenated blood = brain cells start dying irreversibly (cerebral hypoxia)
- CPR generates ~25-30% of normal cardiac output — enough to keep the brain alive
- 1-10-1 Cold Water Rule: ~1 min breathing control, ~10 min useful swimming, up to 1 hr before hypothermia
- Secondary drowning: fluid in lungs can develop HOURS after a near-drowning. Always get medical check.

Activity 2.1 — Physiology Tasks

~15 min

Word Bank: cardiac arrest · submersion · laryngospasm · hypoxia · cerebral hypoxia

Order	Event	What it means
1 (first)		Person goes underwater
2		Airway closes involuntarily
3		Oxygen levels in blood drop
4		Heart stops pumping effectively
5 (last)		Brain cells start dying — time critical!

Q2.1b — Why Start CPR Quickly? [3 marks]

Use the sentence starters to explain the 4-minute window in plain language.

When the heart stops, the brain stops getting _____ blood.

After about _____ minutes, brain cells start _____

CPR generates ~____% of normal cardiac output, which is _____

This is why starting CPR within 4 minutes is important: _____

3 Social Determinants of Drowning Risk

Activity 3.1 — Social Determinants Analysis

Pairs · ~18 min

Word Bank: Gender norms (masculinity) · CALD background · Socioeconomic status · Rural / regional location · Alcohol use

Social Determinant 1: _____

How it
increases risk:

Evidence /
example:

Prevention
strategy:

Social Determinant 2: _____

How it
increases risk:

Evidence /
example:

Prevention
strategy:

Q3.1b — Individual vs Structural [2 marks]

Explain the difference between individual and structural explanations. Give one example of each using the CALD drowning data.

Individual explanation blames _____ choices. Example: "They _____"

Structural explanation focuses on _____ factors. Example: "In many countries of origin, _____"

Structural explanations are more useful for prevention because: _____

4 Evaluating Drowning Prevention Strategies

Activity 4.1 — Prevention Strategy Evaluation

~15 min

Word Bank: Strong — backed by solid research · Moderate — some research, some gaps · Limited — little evidence or limited reach

Strategy	Evidence strength	Key limitation
4-sided pool fencing		
Mandatory swim lessons (children)		
Expanding patrolled beach patrols		
Warning signs at unpatrolled beaches		
CALD community swim programs		

Strategy 1 — \$5M Priority Choice:

I would choose
this because:

The evidence
shows:

The trade-off is:

Strategy 2 — \$5M Priority Choice:

I would choose
this because:

The evidence
shows:

The trade-off is:

5 International Comparison

Activity 5.1 — Research Comparison

~20 min · Devices required

Websites to visit:

royallifesaving.com.au | rlss.org.uk/education | redcross.org/take-a-class/water-safety

Question	RLSSA (Australia)	RLSS UK	Red Cross (USA)
Main program name			
Key safety message / slogan			
One unique feature			

Q5.1a — Should Victoria Adopt It?

Element I
chose:

From (country /
organisation):

Why it could
help in Victoria:

I think Victoria
should / should
not (circle)
adopt it
because:

Source URL:

6 Diving, Spinal Injuries & High-Risk Teenage Behaviour

Word Bank: cervical spine · quadriplegia · spinal cord · sandbank · tidal · sediment · false confidence · masculinity norms · peer pressure · AIHW · RLSSA · feet first · permanent disability · inland waterway

Key Facts — Use These in Your Answers

- Unsafe diving = 10% of all spinal cord injuries and 20% of quadriplegia cases in Australia (AIHW)
- 95% of aquatic spinal injury patients are male; 75%+ are under 30; 66% suffer permanent disability
- Highest incidence of diving-related spinal injury: children aged 10-14
- Royal Life Saving Australia: more people drown in inland waterways than any other location in Australia
- "Feet first, first time, every time" — the core prevention message

Three Australian Case Studies — Read Before Answering

For reference

Case Study 1 — Terrigal Wharf (NSW, 2025)	Case Study 2 — Torquay Front Beach (Victoria, 2025)	Case Study 3 — Bawley Beach Jetty (NSW, 2024)
Teenage boy dived from a wharf at Terrigal into shallow water and suffered serious spinal injuries. Helicopter evacuation to hospital. He did not know how shallow it was beneath the wharf structure.	Teenage boy suffered a serious suspected spinal injury diving at Torquay Front Beach and was airlifted to hospital. Surf beach sandbanks shift with every tide — making depth at any point highly unpredictable.	Young man seriously injured diving from a jetty near Bawley Beach. A sandbank had recently formed beneath the jumping spot, dramatically reducing depth. He had jumped there safely before — false confidence.

Activity 6.1 — Case Study Analysis (Scaffolded)

Individual · ~18 min

Q6.1a — Two common factors across all three case studies

Common factor 1:

How it caused the injury (not just what it was):

Common factor 2:

How this one caused the injury:

Q6.1b — Why 95% of patients are male (structural explanation)

Social norm /
expectation that
drives
risk-taking:

How this leads
to jumping or
diving
behaviour:

Why telling
individuals to
"be careful" is
not enough:

What structural
change would
address this:

Q6.1c — Why neck/cervical spine injuries cause permanent disability

What the
cervical spine
does (function):

What happens
on impact
(physiological
process):

Why this cannot
be reversed
(connect to
Section 2):

Activity 6.2 — Myth vs Evidence (Scaffolded)

Individual · ~15 min

Use the sentence starters below. Replace the underline with your own words.

Myth: "If others are jumping there, it must be safe."

Sentence starter: This is a myth because depth can change due to _____ and _____. A real Australian example is the Bawley Beach case, where a _____ had formed.

Myth: "Clear water means deep water."

Sentence starter: Water clarity tells us _____ about depth. Shallow water can be clear because _____.

Myth: "I've jumped here before — I know this spot."

Sentence starter: Prior experience creates false confidence because _____ changes between visits. The Bawley Beach case shows that a _____ formed between visits.

Myth: "Good swimmers are safe from diving injuries."

Sentence starter: Swimming ability is irrelevant to diving injuries because injury happens on _____ before swimming ability matters. Australian data shows _____ % of patients were healthy young males.

Activity 6.3 — Prevention Strategy (Scaffolded)

My target group
(be specific —
age, gender,
location):

Two statistics
showing this
group is at
higher risk:

My prevention
strategy (what,
channel, who
delivers):

SMART metric
— how I will
know if it
worked:

One limitation
of my strategy:

7 Assessment Task Planning

Choose ONE option. All options need at least 3 sources + 1 statistic.

Option A — Policy Brief (600-800 words): Address Victorian Minister for Sport. Two recommendations + evidence + success metrics.

Option B — Campaign: Target ONE high-risk group. Name, message, visual, platform, 100-word justification.

Option C — School Audit (advanced): Audit school provision against Curriculum 2.0. Priority matrix + recommendations.

My Planning Notes

Option chosen:

Who am I
writing for (audi
ence/addressee
)?

My key
argument /
central claim:

Statistic I will
use (with
source URL):

Source 1 URL:

Source 2 URL:

Source 3 URL:

My draft — write here or attach on a separate page:
